

THE $p = 3$ CASE OF A CONJECTURE OF CHAUHAN, SHUKLA, AND VINAYAK

ABSTRACT. We prove the $p = 3$ case of Conjecture 4.2(ii) of Chauhan, Shukla, and Vinayak on 3-cut complexes of powers of cycles. Within the conjectured $p = 3$ range, the only cases not covered by the theorem of Chauhan and Shukla are $n = 13$ and $n = 14$. By locally reordering, in their explicit facet order, a block of six facets (five of which change position) and a block of ten, we obtain shellings with one and eight spanning facets. Hence

$$\Delta_3(C_{13}^3) \simeq \mathbb{S}^9, \quad \Delta_3(C_{14}^3) \simeq \bigvee^8 \mathbb{S}^{10}.$$

The finite verification is exact. Both reordered blocks are printed in full below; the underlying facet order is the one defined by Chauhan and Shukla, specialized to $p = 3$, and is not reproduced here.

1. MAIN RESULT

For a graph G on vertex set V , the 3-cut complex $\Delta_3(G)$ is the pure simplicial complex whose facets are the sets $V \setminus T$, where $T \subseteq V$ has cardinality three and $G[T]$ is disconnected [1, 2]. Write C_n^p for the p th power of the cycle C_n .

Conjecture 4.2(ii) of Chauhan, Shukla, and Vinayak asserts that for $p \geq 3$ and $n \geq 4p + 1$,

$$(1) \quad \Delta_3(C_n^p) \simeq \bigvee^{(n^2 - 4np - n + 2)/2} \mathbb{S}^{n-4},$$

and, more strongly, that the complex is shellable [2, Conjecture 4.2(ii)]. Chauhan and Shukla proved this for $n \geq 6p - 3$ [3, Theorem 1.4]. At $p = 3$, their result begins at $n = 15$, whereas the conjectured range begins at $n = 13$.

Theorem 1. *Part (ii) of Conjecture 4.2 of Chauhan–Shukla–Vinayak holds for $p = 3$. Thus, for every $n \geq 13$, the complex $\Delta_3(C_n^3)$ is shellable and*

$$\Delta_3(C_n^3) \simeq \bigvee^{(n^2 - 13n + 2)/2} \mathbb{S}^{n-4}.$$

The proof is the two explicit certificates of Section 2, for $n = 13$ and $n = 14$, combined with [3, Theorem 1.4] for $n \geq 15$. The infinite part of the range is thus that theorem's, and [3] is at the time of writing an unrefereed

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preprint; the content contributed here is exactly the two boundary cases. Part (i) of Conjecture 4.2, and the cases $p \geq 4$ of part (ii), are untouched.

The integral homology ranks for the two boundary cases already appear in Table 2 of [2]. The new point is an explicit shelling certificate, which proves shellability and the asserted homotopy types.

2. THE TWO BOUNDARY CERTIFICATES

Let $V_n = \mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n-1\}$ and

$$d_n(a, b) = \min\{|a - b|, n - |a - b|\}.$$

Two vertices are adjacent in C_n^3 exactly when $d_n(a, b) \leq 3$. We encode a facet $F = V_n \setminus T$ by its omitted triple T . A graph on three vertices is disconnected exactly when it has at most one edge; hence the omitted triples are precisely the triples containing at most one pair at circular distance at most three.

For a shelling $F_1, \dots, F_m$, a facet F_j is *spanning* if $\partial F_j \subseteq \bigcup_{i < j} F_i$. The following elementary criterion is convenient for exact verification.

Lemma 2. *Let $T_1, \dots, T_m$ be distinct omitted triples, set $F_j = V_n \setminus T_j$, and, for $j \geq 2$, define*

$$R_j = \{x \in V_n \setminus T_j : T_r = (T_j \setminus \{y\}) \cup \{x\} \text{ for some } r < j \text{ and } y \in T_j\}.$$

Then $F_1, \dots, F_m$ is a shelling order if and only if

$$(2) \quad (T_i \setminus T_j) \cap R_j \neq \emptyset \quad \text{for every } i < j.$$

Moreover, F_j is spanning if and only if $R_j = V_n \setminus T_j$.

Proof. The standard shelling criterion requires, for every $i < j$, an index $r < j$ and a vertex $x \in F_j \setminus F_i$ such that $F_r \cap F_j = F_j \setminus \{x\}$; see [3, Definition 2.2]. Here $F_j \setminus F_i = T_i \setminus T_j$, while

$$F_r \cap F_j = F_j \setminus \{x\} \iff T_r = (T_j \setminus \{y\}) \cup \{x\}$$

for some $y \in T_j$. This gives (2). Applying the same condition to every $x \in F_j$ gives the spanning criterion. $\square$

Let L_n be the order obtained by specializing to $p = 3$ the explicit order of Chauhan and Shukla in Section 3 of [3], culminating in their Definition 3.10. We use only the defining formulas and verify the resulting finite lists directly. This is deliberate: Section 3 of [3] is written under the hypothesis $n \geq 6p - 3$, that is $n \geq 15$ at $p = 3$, so at $n = 13, 14$ its formulas are being evaluated outside the range in which [3] use them, and none of the lemmas proved there is invoked below. That the blocks partition the facet set, that L_n lists every disconnected triple exactly once, and that the exchange condition holds are all checked on the two finite lists themselves. All positions below are one-based.

For $n = 13$, replace positions 112–117 of L_{13} by

$$(3) \quad \begin{aligned} &(\{3, 5, 12\}, \{4, 5, 11\}, \{4, 5, 12\}, \{5, 9, 12\}, \{5, 10, 12\}, \{5, 11, 12\}) \longrightarrow \\ &(\{4, 5, 11\}, \{5, 11, 12\}, \{5, 10, 12\}, \{5, 9, 12\}, \{3, 5, 12\}, \{4, 5, 12\}). \end{aligned}$$

For $n = 14$, replace positions 146–155 of L_{14} by

$$(4) \quad \begin{aligned} &(\{3, 5, 13\}, \{4, 5, 11\}, \{4, 5, 12\}, \{4, 5, 13\}, \{5, 9, 12\}, \{5, 9, 13\}, \\ &\quad \{5, 10, 12\}, \{5, 10, 13\}, \{5, 11, 12\}, \{5, 11, 13\}) \longmapsto \\ &(\{4, 5, 11\}, \{4, 5, 12\}, \{5, 9, 12\}, \{5, 10, 12\}, \{5, 11, 12\}, \{5, 11, 13\}, \\ &\quad \{5, 10, 13\}, \{5, 9, 13\}, \{3, 5, 13\}, \{4, 5, 13\}). \end{aligned}$$

Denote the repaired orders by $\hat{L}_{13}$ and $\hat{L}_{14}$. Both edits are permutations of a block: the facet set, and the set of triples occurring before any position outside the block, are unchanged. Since an internal permutation of a block changes neither the set of predecessors of a position outside the block nor the set of earlier triples seen from it, condition (2) and the spanning criterion can change only at the 6 and 10 edited positions. In particular no facet outside the two windows is affected, and the spanning facets listed in Proposition 3 all lie outside them.

The unrepaired orders fail Lemma 2 only through the following obstruction pairs:

n	j	T_j	earlier witness triples T_i
13	112	$\{3, 5, 12\}$	$\{2, 5, 9\}$
13	114	$\{4, 5, 12\}$	$\{2, 5, 9\}, \{2, 5, 10\}$
14	146	$\{3, 5, 13\}$	$\{2, 5, 9\}$
14	149	$\{4, 5, 13\}$	$\{2, 5, 9\}, \{2, 5, 10\}$

Proposition 3. *The orders $\hat{L}_{13}$ and $\hat{L}_{14}$ are shelling orders for $\Delta_3(C_{13}^3)$ and $\Delta_3(C_{14}^3)$, respectively. Their spanning facets are the complements of the omitted triples*

$$\mathcal{S}_{13} = \{\{4, 10, 12\}\}$$

and

$$\mathcal{S}_{14} = \{\{0, 7, 13\}, \{6, 12, 13\}, \{1, 8, 13\}, \{5, 12, 13\}, \{2, 9, 13\}, \\ \{4, 10, 13\}, \{4, 11, 13\}, \{3, 10, 13\}\}.$$

In particular, the two orders have one and eight spanning facets.

Proof. The verification described in the section *Exact verification* below reconstructs L_n from the defining formulas in [3], checks that the blocks used there form a disjoint partition of the facet set, and applies (3)–(4). It then performs the following exact checks for each $n \in \{13, 14\}$:

- (1) the listed triples are exactly the disconnected triples, without omission or repetition;
- (2) condition (2) holds for every pair $i < j$;
- (3) independently, for each j , every inclusion-maximal member of $\{F_i \cap F_j : i < j\}$ has cardinality $|F_j| - 1$;
- (4) the spanning triples are exactly those displayed above.

The audit is

n	facets	pairs	exch. fail.	purity fail.	spanning
13	169	14,196	0	0	1
14	238	28,203	0	0	8

Lemma 2 now proves the proposition. $\square$

Remark 4. The two spanning counts are not independent data. For a pure shellable d -complex the number of spanning facets equals h_{d+1} , and this in turn equals $|\tilde{\chi}|$; the reduced Euler characteristics of $\Delta_3(C_{13}^3)$ and $\Delta_3(C_{14}^3)$ are -1 and $+8$, as one computes directly from their face numbers, with the sign $(-1)^{n-4}$. Hence once shellability is known the counts 1 and 8 are forced by the two complexes themselves, independently of any order, and they act as a consistency check on the verification rather than as a separate claim. What is not forced is which facets are spanning: that depends on the order.

Proof of Theorem 1. Proposition 3 treats $n = 13, 14$. Since every facet has $n - 3$ vertices, these complexes have dimension $n - 4$. A pure shellable d -complex with β spanning facets is homotopy equivalent to a wedge of β copies of $\mathbb{S}^d$ [4, Theorem 12.3]; see also [3, Theorem 2.3]. This gives $\mathbb{S}^9$ and $\bigvee^8 \mathbb{S}^{10}$ in the two boundary cases. For $n \geq 15$, the result is [3, Theorem 1.4]. Finally,

$$\binom{n-6}{2} - 20 = \frac{n^2 - 13n + 2}{2},$$

so the multiplicity agrees with (1) at $p = 3$. $\square$

EXACT VERIFICATION

Every computation above is finite and exact, carried out in integer arithmetic on subsets of $\mathbb{Z}/n\mathbb{Z}$; no floating-point arithmetic is involved. From the facet criterion of Section 2 one enumerates the 169 and 238 disconnected triples. The order L_n is generated from the defining formulas of [3, Section 3] at $p = 3$, the blocks (3) and (4) are substituted, and condition (2) is then evaluated on all 14,196 and 28,203 pairs $i < j$, together with the purity and spanning tests listed in the proof of Proposition 3. As a control, the same four checks were applied to the order of [3, Section 3] at $n = 15, 16$, inside the range $n \geq 15$ covered by [3, Theorem 1.4], and returned the expected spanning counts 16 and 25. No data file or program accompanies this paper: every quantity quoted above is either printed here or recomputable from the criteria of Section 2 together with the definitions of [3, Section 3], and the program used and its transcript are retained in an auxiliary archive, available on request. What a reader can and cannot check from the data printed here is set out next.

Status. The orders $\hat{L}_{13}$ and $\hat{L}_{14}$ are not printed here. Reconstructing them requires the definitions of [3, Section 3] specialized to $p = 3$; given those, the two blocks (3) and (4) determine the certificates completely. Consequently the following are supported by the computation just described, but are not checkable from this text alone: the zero failure counts of the audit table; the assertion that the four pairs tabulated in Section 2 are the *only* obstructions of L_{13} and L_{14} ; and the identity—as opposed to the number—of the spanning facets in $\mathcal{S}_{13}$ and $\mathcal{S}_{14}$. For the same reason the printed data pin the internal order of each block only up to permutations with the same in-block discharge pattern, and do not pin the identity of the witness triples T_i . What *is* checkable from the printed data is the local part of the argument: that the six and ten listed triples are omitted triples of C_{13}^3 and C_{14}^3 ; that each new block is a permutation of the old one, so that by the locality observation of Section 2 nothing outside the two windows can change; that the four tabulated obstructions are discharged by the reordering, each through the one vertex of $T_i \setminus T_j$ that can ever enter R_j ; that all 15 pairs internal to the $n = 13$ window and 44 of the 45 internal to the $n = 14$ window are discharged from in-block predecessors alone, the exception being the pair at positions 147 and 154, which requires a witness from the unprinted prefix; and, once shellability is granted, the counts 1 and 8, which Remark 4 then forces from the face numbers alone. Shellability itself is not among these checkable items.

Independent recomputation. The computation was re-run independently, by a program written against the statements printed above, and all 138 of its checks agreed. Its reach is narrower than the paper’s in three respects. It does not rebuild $\hat{L}_{13}$ and $\hat{L}_{14}$, which cannot be rebuilt from this text, so the zero failure counts of the audit table and the identity of $\mathcal{S}_{13}$ and $\mathcal{S}_{14}$ as the spanning sets *of those two orders* are not re-established by it. What it does re-establish is the conclusion together with the repair mechanism: shelling orders for $\Delta_3(C_{13}^3)$ and $\Delta_3(C_{14}^3)$, constructed independently and verified on all 14,196 and 28,203 pairs, pure of dimensions 9 and 10, with exactly 1 and 8 spanning facets and realizing $\mathcal{S}_{13}$ and $\mathcal{S}_{14}$; and every assertion of Section 2 that depends only on the printed blocks. Its census of explicitly constructed shellings covers $13 \leq n \leq 32$, its reduced Euler characteristics $13 \leq n \leq 22$, and its homology with $\mathbb{F}_2$ coefficients $13 \leq n \leq 20$; the range $n \geq 15$ of Theorem 1 rests on [3, Theorem 1.4] and not on computation. Odd torsion below the top degree is not excluded by that $\mathbb{F}_2$ computation; it is excluded by shellability together with [4, Theorem 12.3]. A recomputation that agrees is evidence about the computation, not a second proof.

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